Phase 1 seminar document

Submission deadline: 16 October 2026.

Public repository: https://github.com/Black11stV35/econ-disney-seminar

1. Title

An Exploratory Appraisal of a Proposed Disney-Type Theme-Park Complex in Thailand: A Bottom-Up Scenario Analysis.

2. Research question and sub-questions

Primary question. What are the potential tourism and economic impacts of a Disney-type complex in Thailand, and does the project create net value for Thailand once displacement, leakage and public costs are counted?

1. How much incremental international visitor spending would the complex generate, relative to a no-project baseline?
2. Is the complex financially viable for an operator at the documented cost of capital, and what break-even attendance does it need?
3. Does the complex create positive economic net value for Thailand at the national social discount rate?
4. What are the fiscal flows of public participation, and how sensitive are the results to costs, revenue and opening delay?

3. Objectives

1. Estimate incremental international visitor expenditure in Thailand against a no-project baseline, net of displaced spending (H1).
2. Estimate the operator's financial net present value, internal rate of return and break-even attendance (H2).
3. Estimate the national economic net present value after displacement, leakage and resource costs (H3).
4. Estimate the fiscal net flows of public participation as a supplementary check (H4).

4. Hypotheses

- **H1.** The complex generates positive incremental international visitor expenditure in Thailand relative to the no-project scenario, after spending displaced from other Thai destinations is subtracted.
- **H2.** The complex has a positive financial net present value under the documented Thai base-case demand and cost assumptions.
- **H3.** The complex produces a positive economic net present value for Thailand after incremental resource costs, displaced activity and material social costs are accounted for.
- **H4 (supplementary).** The discounted incremental tax and fee revenue attributable to the project exceeds its discounted project-specific public expenditure and fiscal costs under the stated policy scenario.

Decision criteria. H1 holds if the base-case net incremental spending is positive after displacement. H2 holds if the financial net present value is positive at the financial cost of capital (WACC of 8 percent in the

base case, recommended; see section 7.3). H3 holds if the economic net present value is positive at the social discount rate of 7 percent (NESDC 2024). H4 holds if the fiscal net present value is positive under the stated policy scenario.

5. Scope and scenarios

Scope. The primary appraisal covers the THB 300 billion sports and entertainment complex, the high-expectation scope of the Thai proposals. The park-only concepts of THB 100 to 200 billion are different projects with their own demand, revenue and cost boundaries. They are reported as separate cases only if time allows.

Scenarios. A base case, low and high demand cases, and the sensitivity cases of section 7.5. The sensitivity cases follow the NESDC (2024) guide: construction cost, operating cost and revenue each varied by 5, 10 and 20 percent, a one-year opening delay, and two-factor cases of cost and revenue together.

6. Literature

(a) Appraisal standards. The ADB (2017) guidelines set a minimum required economic internal rate of return of 9 percent for investment projects and 6 percent for social sector and environmental projects (paragraph 194), and require sensitivity analysis and switching values (Section VII). The NESDC (2024) guide recommends a social discount rate of 7 percent for appraisals begun in 2025 or later, within a range of 6.55 to 7.59 percent; the rate of 12 percent applies to appraisals begun before 2025. It sets a standard conversion factor of 0.7 for converting financial costs to economic costs where shadow prices cannot be computed.

(b) Theme-park and single-attraction evidence. Plaza (2006) estimates the effect of the Guggenheim Museum Bilbao on overnight stays, jobs and the return on investment. The published article reports 61,742 additional overnight stays a month from a single-series time-series model with no control group, so it supports context rather than a causal transfer. Published estimates for other attractions are reviewed in the causal benchmark note and are not transferred to Thailand (section 7.6).

(c) Historical cases. Hong Kong Disneyland and Shanghai Disney Resort are the closest cases. Their evidence is used for calibration ranges and as context, not as a transfer coefficient (section 7.6). Chan (2004) studies residents' perceptions of Hong Kong Disneyland and its impacts. The thesis reports two telephone surveys of 1,060 respondents and is verified against the PolyU repository record. Two Henderson studies cover Southeast Asian destinations. Henderson (2012) studies the development and regulation of casinos in Singapore. It reports visitor arrivals of 11.6 million in 2010 and a 50 percent rise in tourism receipts, and it notes that the 2010 upturn cannot be attributed solely to the integrated resorts, because the 2008 and 2009 uncertainties also exerted a strong influence. Henderson et al. (2013) review LEGOLAND Malaysia, the first LEGOLAND in Asia, which opened in Johor in 2012. The article states that its findings rest on public-domain material and two visits to the park. It reports that Universal Studios Singapore had about 3.4 million visitors in its first year (TEA/AECOM, 2012), that management expected one million visitors a year in the early years (Malaysia Today, 2008), and that the General Manager said full recovery of the investment usually takes 15 to 20 years (Lee, 2012). These are press-sourced statements. They are used as context and are not inputs to the model.

Gap. The literature provides single-attraction, macroeconomic and appraisal-standard evidence, but no bottom-up model for a Thai complex that follows the national appraisal guidance. This study fills that gap.

7. Method

7.1 Framework. A bottom-up appraisal that compares the with-project and without-project situations, following the ADB (2017) guidelines and the NESDC (2024) guide.

7.2 Demand, spending and infrastructure. The working hypothesis is that public infrastructure variation is the main driver of the tourism effect. The model therefore caps incremental visits by the infrastructure that can serve them, namely airport headroom and accommodation supply. Annual passenger movements convert to visits with `visits_from_passengers`, using two movements per round-trip visit, which is an assumption to be confirmed. Incremental visits are capped by headroom in `capped_net_spend`, and net spending follows from the capped visits after displacement.

Sourced values in the input table include U-Tapao terminal capacity of 3.7 million passengers a year (EEC Office, page status October 2025); the AOT six-airport total of 124,033,786 passengers in calendar 2024 and 100,058,695 in fiscal 2023; Suvarnabhumi and Don Mueang traffic for 2019 and 2024 (AOT); Chon Buri occupancy of 80.42 percent in 2024 (National Statistical Office, Table 17.1); Chon Buri foreign visitors of 11,689,827 in 2024 (TAT Intelligence Center); the national arrivals of 2020 to 2023 and of 2025 and the tourism receipts of 2024 and 2025 (Ministry of Tourism and Sports and National Statistical Office); the length of stay of international visitors in 2016, 2017 and 2024 (National Statistical Office and Ministry of Tourism and Sports); the attendance and financial figures of Hong Kong Disneyland and Ocean Park, used as comparators; the Bank of Thailand exchange rate; and the Revenue Department tax rates. Hotel and guesthouse rooms of 956,372 refer to calendar 2021 (NSO Statistical Yearbook Thailand 2024, Table 17.4). Pattaya hotel keys of 31,581 in H2 2024 and 34,743 in H2 2025 are CBRE broker data, not an official source; the 2024 figure is used as a proxy for the room stock of the complex.

Team decisions, confirmed by the leader on 14 October 2026 and recorded as labelled assumptions in the assumptions table: WACC of 8 percent in the base case (10 and 7 percent in the low and high cases), and a project life of 20 years in the base case (20 and 25 years in the low and high cases). Attendance, incremental and displacement shares, domestic retention, the spending-per-visit base and high cases and the 85 percent occupancy ceiling in the base case keep their working values. The spending-per-visit low case is a sourced on-site benchmark, converted at the Bank of Thailand yen rate of 12 October 2026. Apart from that, none is a sourced value. Inputs that could not be sourced are blank in the input table, and figures that were not confirmed are flagged unconfirmed, never sourced. The base case stops only when a sourced or proposal input is pending; team decisions do not stop it.

7.3 Three ledgers.

- **Operator ledger:** revenue minus operating costs and capital expenditure, giving the financial net present value, internal rate of return and break-even attendance at the financial cost of capital (WACC of 8 percent and a project life of 20 years in the base case, as recommended; the NESDC guide requires both to be stated and justified). Operating costs include depreciation, by the leader's decision.
- **Fiscal ledger:** tax and fee revenue from the net spending minus public capital and operating costs.
- **National economic ledger:** economic benefits minus economic costs, discounted at 7 percent. Economic costs use the conversion factor of 0.7 unless shadow prices are computed. The economic benefit is the net spending of the incremental visits, following ADB (2017) paragraph 146, which counts the net increase in local incomes from incremental tourist expenditure. The whole operating cost of the complex is charged.

7.4 Macroeconomic checks. The NESDC guide asks large projects to report wider effects, including the balance of payments and labour market effects. These are reported as checks on the results.

7.5 Sensitivity and risk. Single-factor cases of 5, 10 and 20 percent on construction cost, operating cost and revenue, a one-year opening delay, two-factor cases of cost and revenue together, and switching values for each key assumption, following ADB (2017) Section VII. The sensitivities are applied to the parameters behind the aggregates, not as mechanical percentage changes.

7.6 Use of historical evidence. Published estimates are not transferred to Thailand. The causal benchmark note shows that no causal study of a Disney-type opening exists that could be carried over, and that gross impact studies have no counterfactual. Thai inputs are used wherever they are sourced. Where no Thai value exists, the Hong Kong Disneyland and Ocean Park figures are used as labelled comparables for attendance, visitor mix, ticket price and cost ratios (generalized in the assumptions table). Historical cases inform the ranges and their limits are reported.

7.7 Reproducibility. The model is in the project repository (`disney-tourism-thailand/src/dtt/appraisal.py`), with its inputs in `data/appraisal/inputs.csv`. Each row gives its value, unit, source and status. The base case stops only when a sourced or proposal input is pending. The unit tests and repository checks pass.

8. Decision rule

The project is supported as a basis for decisions if H3 holds at the 7 percent social discount rate, and if its sign is unchanged under the ±10 percent cost and revenue cases and the one-year delay. Where the sign changes, the result is reported with the conditions under which the project would pass. Break-even attendance is reported for the operator ledger in every case.

9. Limitations

- Most Thai demand, spending and operating inputs are not sourced. Several are labelled generalized or assumed values in the assumptions table, and the base case runs on them. Its results are scenario ranges, not sourced estimates.
- AOT traffic for calendar 2024 and for 2019 at Suvarnabhumi and Don Mueang is sourced. U-Tapao's calendar 2024 passenger count of 384,881 is derived from 24 monthly records of the Ministry open-data file; it has no sourced calendar 2019 figure, because AOT does not operate it. Chon Buri occupancy is from National Statistical Office Table 17.1; its 2019 value is unconfirmed. The base occupancy ceiling of 85 percent is an assumption.
- Sources conflict on the high-speed rail opening year (2031 from the EEC Office; Wikipedia gives 2029 in its infobox and 2030 in its own later status entry). The model does not use the rail date. The Chon Buri figure of 28,156,302 is a Thai plus foreign total and is not used as foreign arrivals; the foreign visitor count is 11,689,827.
- The proposal has no signed agreement, approved budget, named operator or site allocation.
- The historical evidence does not identify the Disney-specific effect separately from concurrent policy changes.
- The output is a set of scenario ranges, not confidence intervals.
- Several comparator anchors rest on secondary copies: the Ocean Park attendance for FY2018/19 (a search summary of the results announcement), the Lotte World ticket price for 2026 (a news report) and the Shanghai Disneyland attendance for 2019 (a copy of an index). The Ocean Park operating costs for FY2018/19 come from its annual report.

10. References

- Asian Development Bank. (2017). *Guidelines for the economic analysis of projects*. Manila: ADB. https://www.adb.org/sites/default/files/institutional-document/32256/economic-analysis-projects.pdf. DOI: http://dx.doi.org/10.22617/TIM178607-2
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- Plaza, B. (2006). The return on investment of the Guggenheim Museum Bilbao. *International Journal of Urban and Regional Research*, 30(2), 452-467.

Appendix A. Supplementary historical analysis: record of results and limitations

Date of record: 16 October 2026 (UTC+7).

Repository: `econ-disney-seminar` (public), folder `disney-tourism-thailand`.

Code commit `0197a1f` (notebook 05 and its test), inputs commit `15a5ff1`, results commit `a811b0b`. These commits are in the history of `main`. The first run that this record reported, of code commit `b6bb77f` and results commit `e206a5b`, is superseded by the re-run described in section 2.

1. Status

The analysis runs on the dataset only. The clean run of 16 October 2026 completed every step, except the SkylakeX kernel, which cannot run on the computer used for the run. The route rule selects the ambient route, and the primary estimator is A2. The primary effect for the proposal of 300 billion baht has a median of 0.40 percentage points of GDP, with a 5 to 95 percent range from -1.14 to 1.94 points. At the 2025 baseline that is 2.33 billion US dollars a year (range -6.55 to 11.17 billion). The weighted transport gives a point estimate of -0.73 points, with the opposite sign. Both are reported. This record reflects the re-run of 16 October 2026, which uses the A2 interval for a new study and the placebo nulls without rescaling. The primary effect and the transport results are unchanged by it; the interval for a new study and the detection figures change.

2. Run provenance

- Run record: the status, summary, manifest, package list, interpreter version, World Bank check and test totals of the first run are in `seminar/provenance/run_2026-10-16/`. That run started at 08:12:45 and finished at 10:01:05 on 16 October 2026 (UTC+7). The re-run that this record reports started at 21:13:13 and finished at 22:19:51 on 16 October 2026 (UTC+7). Its package list, interpreter version, timings and final test totals are in `seminar/provenance/rerun_2026-10-16/`.
- Code of the run: the pipeline files of commit `0197a1f` are identical to the files that ran, because the code was committed after the re-run without changes. The notebooks ran with `DTT_RUN_REAL=1` on the project virtual environment, with the default kernel. The notebook times of the re-run were 00 19 s,

01 34 s, 02 52 s, 03 3778 s, 04 69 s and 05 45 s, 3997 s in total. The outputs of notebooks 00 to 04 are unchanged in content; notebook 05 writes the two new outputs. The first run, of commit `b6bb77f`, took 4711.5 s; its notebook times were 00 11.6 s, 01 23.6 s, 02 36.0 s, 03 4511.4 s, 04 55.9 s and 05 28.0 s.

- Manifest of the first run: `manifest_sha256.csv` was taken at 08:13, before the first step, and lists 160 files. Its entries for `results_20261016\`, `figures_20261016\` and `executed_20261016\` are outputs of an earlier run. They are not part of the final tree.
- For the first run, text changes after the manifest was taken in `README.md` (World Bank status, kernel statement, result section and thermal sentence), the first markdown cell of notebooks 00 to 05 (run-time paragraph only), and the heading list in `tests/test_notebooks_static.py`. No code cell changed after the manifest was taken.
- Python 3.12.8 (MSC v.1942, 64-bit) on Windows 11 Home. Package versions from `pip_freeze.txt`: cvxpy 1.9.3, matplotlib 3.11.2, numpy 2.5.4, pandas 3.0.6, POT 0.9.7.post1, pytest 9.1.1, scikit-learn 1.9.1, scipy 1.18.1, statsmodels 0.15.0. The re-run used the same versions in the project virtual environment, which was built from `requirements.txt` with POT 0.9.7.post1.

3. Results of the run

Route rule (`results/route_decision.json`, rule hash 852b46da9afa092222650d1709049c85d02397fe5f83d3b46eb5e68c9e368b7f):

- Importance model not usable. The primary sample has 19 episodes, below the minimum of 20. The cross-validated R-squared is 0.072 (minimum 0.10), the permutation p-value is 0.436 (maximum 0.10) and the split-half correlation is -0.148 (minimum 0.50).
- Leave-one-economy-out RMSE: weighted transport 1.015, plain mean 1.134, uniform transport 1.006. The ratio of the weighted transport to the uniform transport, 1.009, is above the limit of 0.95, so this criterion fails.
- Target supported: effective number of sources 6.75 (minimum 2), largest source share 0.25, and no material feature outside the range of the sources.
- Route: ambient. Estimator: A2, a scaling law through the origin in the cost share, with LOEO RMSE 1.038 against 1.138 for A1 (ratio 0.913, below the limit of 0.95) and 1.630 for A3. The nearest-neighbour predictor nn3 has the lowest LOEO RMSE of all, 0.903. The rule does not use it.

Scenario complex_total (proposal of THB 300 billion, cost year 2025, 1.58 percent of GDP, US$ 9.12 billion at 32.8828 THB per US dollar, receipts 12.15 percent of GDP on the panel basis):

- Primary effect, percentage points of GDP: median 0.40, 5 to 95 percent -1.14 to 1.94, 25 to 75 percent -0.21 to 1.01.
- Primary effect, US$ billion a year: 2.33 at the 2025 baseline (-6.55 to 11.17) and 2.19 at the 2019 baseline (-6.18 to 10.53).
- Primary effect, percent of the counterfactual receipts: median 9.96, 5 to 95 percent -69.87 to 89.43.
- 90 percent interval for a new study, from the leave-one-economy-out errors of the primary estimator A2: -1.52 to 2.33 points, centred on the point prediction of 0.40 (half width 1.92, from 19 cases). The weighted-transport interval of notebook 04, a diagnostic, is -2.73 to 1.27 points (half width 2.00). Minimum detectable effect at alpha 0.05 and power 0.8, with the placebo null used without rescaling: 2.00 points (normal approximation 2.23). Detection probability over the effect draws: 0.19.
- Weighted transport, a diagnostic and not the primary route: point estimate -0.73 points. Sensitivity median -0.71 points, 5 to 95 percent -1.28 to -0.19.
- Scenarios of 200 billion and 100 billion baht: primary medians 0.27 and 0.13 points.

- The government claim of about 1 percent of GDP a year (February 2026, no study cited) appears only as a reference scale.

4. Verification

The rows for the first run (code commit `b6bb77f`) were not repeated for the re-run, except where a row says so. The last row is the full suite on the code of the re-run.

Run	Result
pytest, default kernel, dataset check skipped (`pytest_default.log`)	1795 passed, 1 skipped
pytest with `DTT_RUN_REAL=1`, including the dataset check (`pytest_dataset.log`)	1796 passed
pytest with OpenBLAS kernel Haswell (`pytest_Haswell.log`)	1795 passed, 1 skipped
pytest with OpenBLAS kernel Zen (`pytest_Zen.log`)	1795 passed, 1 skipped
pytest with OpenBLAS kernel SkylakeX (`pytest_SkylakeX.log`)	Process stopped with an illegal instruction, exit code -1073741795, after 25.8 seconds
Dataset chain, notebooks 00 to 05 in order, first run (`dataset_chain.log`)	Exit 0, 4711.5 seconds
Dataset chain, notebooks 00 to 05 in order, re-run of 16 October 2026 (`chain.log`)	Exit 0, 3997 seconds
Notebook 01, comparison with the reference values	1179 of 1216 pass; 37 differ
World Bank check against the live API (`wdi.log`)	96 economies, 12,868 compared cells, 12,868 equal, 0 different, 0 present in only one source, 1,532 empty in both
Code commit `b6bb77f`, full suite including the appraisal tests, first run	1877 passed, 1 skipped
Code commit `0197a1f` with inputs commit `15a5ff1`, full suite of the re-run (`pytest_final.log`)	1879 passed, 1 skipped

The thermal log `results\thermal_log.csv` of the re-run has 763 rows. Its temperature column is empty in every row, so no CPU temperature was recorded. Every row shows the action `ok`, and the guard relied on the CPU load. The first run's log had 196 rows with the same pattern.

5. Limitations and disclosures

1. Sample size. The primary sample has 19 episodes, one below the 20 that the importance verdict requires. Under the fixed rule this sends the analysis to the ambient route. That is the outcome of the rule. The importance model is not usable on any of its criteria, and under five seeds the verdict was usable in none of them. The variant with all 27 feasible cases also fails.
2. Close choice of estimator. The selection of A2 over A1 is flagged as close, because its RMSE ratio of 0.913 lies within 0.05 of the limit of 0.95.
3. A predictor with lower error is not used. The nearest-neighbour predictor nn3 has an LOEO RMSE of 0.903, below 1.038 for A2 and 1.015 for the weighted transport. The rule does not use it, and it is reported here.

4. Sign disagreement. The primary median is +0.40 points of GDP, and the weighted transport point estimate is -0.73 points. The weighted transport value is the same at the 200 billion and the 300 billion baht scenarios, so the disagreement holds at every cost level in the scenario table.
5. Precision. The predictive interval includes zero. The 90 percent interval for a new study, from the errors of A2, is -1.52 to 2.33 points, about 3.9 points wide. The minimum detectable effect of 2.00 points is about five times the point estimate of 0.40. The detection probability at the headline is 0.19, which is not the power at the minimum detectable effect.
6. Placebo noise. The placebo nulls are used without rescaling. The placebo effects come from the source economies, whose receipts share is 4.36 percent of GDP, against 12.15 percent for the target. If placebo noise grows with the receipts share, the detection threshold and the minimum detectable effect understate the noise at the target.
7. Scenario widths. Only the medians scale with the cost. The interquartile width of the primary effect is 1.22 points at 300 billion baht, 1.18 at 200 billion and 1.15 at 100 billion.
8. Cost year. At the 2019 cost year the capital cost is 1.78 percent of GDP, about 12 percent above the 1.58 percent at the 2025 cost year.
9. Baseline years. Receipts fall by about 24.5 percent between the 2019 and 2025 baselines, which affects the relative results.
10. Derived costs. The file `data/processed/cases_investment_fx.csv` holds 27 dollar costs. Twenty are converted from local-currency figures at official annual-average exchange rates, and seven are reported in dollars. All 27 are derived estimates, not reported costs, and 15 have an assumed price year. Costs are known for 28 of the 44 episodes only with the derived dollar costs, and for 12 without them.
11. Proposal status. The THB 300 billion figure is a proposal. No signed agreement, approved budget, named operator or site allocation was found in the sources reviewed.
12. Calibration. The 90 percent interval of A2 was checked on the 19 episodes of the dataset by leave-one-out. The half width set from the other 18 episodes contains the held-out error in 18 of 19 cases (94.7 percent). The one case outside is MAR_2009_ep, with an error of 2.78 points against a half width of 1.92. With 19 episodes this check is weak evidence on its own.
13. Kernels. SkylakeX cannot run on the computer used for the run. Haswell and Zen agree with the default kernel on every test they run. The check on the dataset is skipped in those runs unless `DTT_RUN_REAL=1` is set, and it passed in the dataset run.
14. World Bank files. The 20 files in `data/raw/wdi/` were retrieved with a web fetch tool during the project, not by a direct download from the World Bank API. The comparison with the live API on 16 October 2026 found no difference in any of the 12,868 compared cells.

6. Open items

- No item in this record is open.

Annex 1. Source files and SHA-256 values of the final tree

Values are the SHA-256 of the committed content of each file in commit `a811b0b`, folder `disney-tourism-thailand`, as listed by `git ls-tree -r`. Paths are relative to the folder `disney-tourism-thailand`.

File	SHA-256
`.gitattributes`	`12fb904caa0ce0b078965bcbe9bb44b4635c31b0e4eb9bc6f17db2a970b63b7e`

File	SHA-256
.gitignore	652a83d3587e05bc17d24ba840792bd508393a5929f5824db856968d504548d7
DATA_NOTICE.md	8b59d675e36a7dc49f1c11d5053a13d2f796482d686d9b083ed531caa49b2d6e
README.md	2ad63fa76a5073b7a26bc410c2d3bd9e2e4ec061dbafdc05a8064f716724d4bc
data/appraisal/assumptions.csv	65dffd1b284565e57cde589e6799166d43fdfefff0300749c174653aad9568b5
data/appraisal/inputs.csv	c3ed3817939f80f6700cf6bafa03f13bb9dcdb2d92de93d9970c05567d50ac3e
data/processed/cases_catalogue.csv	cd3d24b78aa8f1ca380bdd24e89bfdf9ecd8d687c30a375ce78c69ad6a068b91
data/processed/cases_investment_fx.csv	5567ea28efd9511a7b999ec558923ac78daa8a7738ab5a28c16c35c6658258f7
data/processed/thailand_baseline.csv	2a5b4d28baea34fd6a37ba54a73c8aade4d29ad55868cec2c59f81817d6ab534
data/processed/thailand_proposal.json	4075784fbcf00c96109050f84701b0121364fcc7bde0693b142d437188edc860
data/raw/disney_did_panel.csv	0c9dd53828ad81503079f1b1a63ba5c7e15de5b0e313ccc99a439bce6a9fc46d
data/raw/wdi/country_metadata.csv	319d8c248ba5386c110be64ee2915d7b11a892f5d90ae1b1f9df97f8bab3e1e6
data/raw/wdi/fetch_log_G1.json	3f6a84457b48b8898dadf7d7229ae25e437575e101b078fe3bd68f940ccede4b
data/raw/wdi/fetch_log_G2.json	502336576c945d011c44733b2b66f1ff897ebd4b8c84d2001a6eedc3647d3cba
data/raw/wdi/fetch_log_G3.json	7b66d00c143dcaeccb30a34187097af3d846a30dd02a75b7de4887ed7649f13d
data/raw/wdi/fetch_log_G4.json	7926977509215fb4f11c0b41abcf30f4099890a551f16be89d29f4ee9f67f064
data/raw/wdi/wdi_G1_IS.AIR.PSGR.csv	7b0815d0d813e1ec50a38a876aba1e2a381653eb545c3f5f3a2209efa07e8e3f
data/raw/wdi/wdi_G1_NY.GDP.MKTP.CD.csv	7d435f99b7fe4673a639769e2e060a3cd4f2f699714b2063db50b4adf6c201de
data/raw/wdi/wdi_G1_SP.POP.TOTL.csv	8812e42d470e39b8ca664179d88f20afaf41f1a67d05eca929c1cf0e99d85bbc
data/raw/wdi/wdi_G1_ST.INT.ARVL.csv	09c776e55878f46fada16dd431f1cb066ad863e5a84650d90bbcf07b32440500
data/raw/wdi/wdi_G1_ST.INT.RCPT.CD.csv	24df255f656c78f2ccc413aa45083deabebbcba300334e44c9f0ee628ef747f3
data/raw/wdi/wdi_G2_IS.AIR.PSGR.csv	5c24f1f46b8958382101e5819455f08db27560c60600a1469b4a460a0cf086e2

File	SHA-256
data/raw/wdi/wdi_G2_NY.GDP.MKTP.CD.csv	543bfe7d3c62727bf8935ad619813b0b1f5283edf5f0c593efcf4a2b4524f6ad
data/raw/wdi/wdi_G2_SP.POP.TOTL.csv	67d02d94fff68ba6e5f646a7484c34c46bad0e20d853d96fa4d040b13f6bd88b
data/raw/wdi/wdi_G2_ST.INT.ARVL.csv	8effd8fbd7ae171b43fca0d464f58ff205f7c8959776ff21ba986b54c70d2dca
data/raw/wdi/wdi_G2_ST.INT.RCPT.CD.csv	5ef445907bfc509bf1d2f7257fc70b3cd6d483e6e554cbea64a9def999f5d06b
data/raw/wdi/wdi_G3_IS.AIR.PSGR.csv	54dca09a19b896dd6c35d9558cfadcf796c5c2a30c239b83a5bffdf3db04e16b
data/raw/wdi/wdi_G3_NY.GDP.MKTP.CD.csv	5e2ea5e9ed1c624601195252c4a86d5e934cc93a13b8efd72766dcd70c5dfe9e
data/raw/wdi/wdi_G3_SP.POP.TOTL.csv	c4e13a5c9c30cd76549666cec51787fa3369b8ff81d953304d5ac6ad69316d9b
data/raw/wdi/wdi_G3_ST.INT.ARVL.csv	74793ff52e334f21e55efec1fd71c1a14a964a423486bd324a26ed8f856e0b5a
data/raw/wdi/wdi_G3_ST.INT.RCPT.CD.csv	486221b593ce4e6ba47fa5c6f315b7e6ff7311709e5cd9589ed806834c791df6
data/raw/wdi/wdi_G4_IS.AIR.PSGR.csv	fc48adec44929bde9f299ec1d229271264c1cf4bf9c69318f4f8e41fea1ee615
data/raw/wdi/wdi_G4_NY.GDP.MKTP.CD.csv	953b1d536bf38a8131a2673fe47500d365f3f39fa2cc9b15f4f234c3722a227b
data/raw/wdi/wdi_G4_SP.POP.TOTL.csv	625122da7d288112e274f0991d1b4cf9f7e6b7823b4b4c28a8e68fcb4ee18ec0
data/raw/wdi/wdi_G4_ST.INT.ARVL.csv	1166dd6909561f5a865cdd15ab77bd5c27df23c5d6f60d54ca8d1bb8d3fdc555
data/raw/wdi/wdi_G4_ST.INT.RCPT.CD.csv	737994c36b7379a31ac98129aec1dff98dadb434d051e2d29fd6baa18dc89d30
data/reference/replication_reference_values.json	7b0c792437d210aa49ffe7cd4733dea27d5cdb722745cfad6fb789dc7f7107a4
docs/cases_catalogue_notes.md	0232abdc88f6d81add7b944df5429c2392d0e998e8cc1b693e2ce714b792e542
docs/causal_benchmark_search.md	3798f920a3378343ee7788bce45067ac8996786a073e7d42efaff3f98549da22
docs/investment_fx_notes.md	8dfd07f22de3be5b45031a4f1c7108dafc14a754a4dea8bf5f0cc2c3ef174cab
docs/thailand_baseline_notes.md	df676ec2caa067232ba8ea7c5761e2650c850b57038e6e16867b30413e9b4d3b
figures/case_effect_vs_capex.png	18c3d03f96d73dabad42538c6560c4e5684b2a4faa1007fc16cf5286e34690a9
figures/case_effects_forest.png	3ebd33db6c78552d46b00b9e5e3bb04bba7ad85038d6ae40dd421b271ffccb02

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figures/case_estimator_comparison.png	7b1821013d8ed7abbad792b6455bdef6f1a688f04d6e38c14fc05dcaaa311663
figures/case_gap_paths.png	d8cdb43c23ba8c98d00d72a9389ead773bd38df115c86ead8c001408a88694ce
figures/impact_routes.png	eee97b844be7af26f6d29076417127f81f3740b51768ade2bc88b885b72f0baa
figures/impact_scaling_law.png	11e551dc12862a1cbd013e7ca5d9b7e0ffedd0537092644d90532ae14c51b0f2
figures/importance_bars.png	b0d54c5723ff428b96f65cc76207aa2f4b7e097a4b41ff89424e0d3c14d62c3a
figures/importance_penalty_path.png	b45a544a6a68648b749aef58ab388aa4e435b0ddd7318da41f626b2f5e8211ec
figures/importance_rank_correlation.png	32634f0402cb11e84679d0ec49130567369cffb324415b2a3fb553938410973c
figures/importance_selection_frequency.png	a32481c0826efd68bb593b5c914c74577aa7df51e7ed323045ec3752620fe7a4
figures/ot_feature_space.png	3df482e1d28219b467e548eed4392c92a5f15a876dbfdfb5b3d92c4e23430243
figures/ot_loco_validation.png	e27a34cd6e4ca331a86af25957b3f401ce066eed03a1c8e0b090617b984b3487
figures/ot_plan_heatmap.png	27be0a3021947e20597ba12060756955a4a143b3bb11b6a00a7b1e5ff4cdfd43
figures/ot_validation_null.png	2a51288f763c59bd7674a8f1eafc77a561e0979bcaeaca253ee5161d6e4a8969
figures/outcome_by_year.png	4e0a9fa14996974188c63b31ecdb2b8a6e41c71bb5d2775f1e23bd25aa0fdcd2
figures/outcome_coverage_heatmap.png	21a574a06df293667ae6e6070f9e6f4e4f104b3c24f7a7b55a489770cdb92fb6
figures/real_event_study_hk.png	e2876940b14bf42a96e60ea39447d3596e2d4a08eee38afa5ca596c398ca0957
figures/real_event_study_sh.png	fdd8018fac891dac92f1d9909b9dc3ed5f82b53945ba4dc067df6110047afbdb
figures/real_placebo_hk.png	70a292dd42898cf9b772d02588afa1b2e5b39ad6ec9cad09984e048431b2b28c
figures/real_scm_hk.png	233a6476c5258d26f72c9d24262595c73d1e649823bc0492a5299ee8ed001375
figures/real_scm_sh.png	1e031e330b638e12b80813fe99e975fe51b9cbe7fe1f4c69e270491b2120b0fc
figures/ref_event_study_hk.png	128847ceff92f764124034c023c25f1e415ce47962ceafc63cd710ac0d20f0ec
figures/ref_event_study_sh.png	146a93b1898e49aa79f97c82cd3629ecb1cbd0fcbc30447e52ed7813387e8e37

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figures/ref_placebo_hk.png	bc8e13a2ffb66c7e0f2fa646376abd614822ab67823dfd2b5a6096c769decbfc
notebooks/00_build_global_panel.ipynb	fc9600af621d9dbf89cba1ba866c6b5cacdb414ca47c603b4d44b471fbb03f88
notebooks/01_replication_and_validation.ipynb	f1a92772cad40c5e1ac8a1e799f900b35707753d0cdb77bb4dde53aaae7781a7
notebooks/02_case_effects.ipynb	93d5483a62038f60a3f7c9bec3e47114fdcf77df43e4543522588a6a34285b1d
notebooks/03_feature_importance.ipynb	3d492e6ffcf14139e19fcef85276bf14e89ee3d3934dd272372407c631ea5d6a
notebooks/04_optimal_transport.ipynb	2dd8b6e22b19ee3ec791e32326debc8da9fc1b78e4141be688f3051ae7fc265f
notebooks/05_thailand_impact.ipynb	39adef7db4cc8d5c40336a77ae4ae2fe8654a8b40c72a0b8825eb89455428ce0
requirements.txt	d10ee69f9aaf3dcff762b6e1df489d51aafb4d7738d3bd0d50ae49f0281449bd
results/ambient_loeo.csv	d6ce302f2bc71e10f68d0c3c33d0d1bba7edad5a80555d849ff3f7c2b5803059
results/ambient_loeo_errors.csv	89fd590cb56b01613a6c5ff21fc92b9bddcd227c4b9cb8ead892526116cf4d1c
results/ambient_new_study.json	52d697a6b4b88909351b4fa87b65cca4b23345babb4d5948698e30444674b42b
results/case_effects.csv	489a4d07127eb78e1ff9ec75c27dfbdffa3aadbfa001b15f26084b7bb2ad70bd
results/case_effects_robustness.csv	0788caee2f9bb4d7758426bc70a26d62c3b46d14de6324b4e8a80c06eaefb067
results/case_features.csv	c94fa5cff06e1532891925cfe47328397e749dc5a429110e3b7f37185d8e51ec
results/case_features_imputed.csv	92facc85bb953c1fb2b95b78597965dc30625e8404bffa1b94e8c3fadfc49472
results/case_investment_fits.csv	1ee29370210040d92b2fe6a45562c792047cebee0e2eb319f15d425e74b3755e
results/case_placebos.csv	c75081aa79277183db4b6fab16395771077639219b56a52397fe565121e94207
results/donor_eligibility.csv	4554e9fd6a6f01c48656a061bf55983d6f7488e5c4a8c2d2069be0459ed443ff
results/episodes.csv	d942ef2fe5bb5d126fc41528da030b4c57a75f735b2eff646ebec98c93e135ef
results/feasibility.csv	112db87394a14e4cb313305cf5a074adb8325ec2837e56e44d1f5bc7f644589c
results/feature_importance.csv	543d91b84f4310f93d84818e66a60563a909a4a15727eb170eb8c4a0ef4a876c

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results/figure_inventory_00_build_global_panel.csv	ec1e929e81bc50f4b466382b055b1f963c4ebd6f1acd91d42a5c5ef6983b2f68
results/figure_inventory_02_case_effects.csv	cc0a1dad3815a3f6cbba23b5c8b45b9497c89710ae062d5f9ecfafd95b20d989
results/figure_inventory_03_feature_importance.csv	e28ee274ecc1335bcf3f96c168d671f186af1750d6885466710001418a88aa26
results/figure_inventory_04_optimal_transport.csv	cad79a17db63475735eff21812604ebdcb5ef6cbf078a335e2047d9adad122b2
results/figure_inventory_05_thailand_impact.csv	6df029faf773a53122cc6c66e7c7000a2b04f5e9a6b8124ba090e11ff63579fe
results/importance_diagnostics.json	8e3ae05daf733c319f4d724c1d983fd2b5f0865ac5e66d4cd91ce205f99a1331
results/imputation_report.csv	1dd00c253bd0c68957522e3fe9856df80cc3f57010890586672650a9b330d2b6
results/loco_fold_weights.csv	a9ae0b87e6adb3f9da64bd01a934a7be22713a02d58c816dfe11ab6662bd0aad
results/loco_ratios.csv	d5c84486144eaf5d38efd1ff2b7a50ed61a15a457aa7c46ce5bfe342aa5689ba
results/loco_ratios_rel.csv	380a975ebb17c362ab8c0516f2529086ff25c5343f4f58de45e5c73d1dea21f4
results/loco_summary.csv	e930aa2454a7174cb36c676bd1b43929c7ce4fe52f97e5d50f33b0df11492502
results/loco_summary_rel.csv	57f46821f2180b6c5bb666600d1f43a5603de4c74edc33986e46ceb3c958fb5f
results/loco_table.csv	decbbf86725d6e099e250ef3cb331784c8b17f648fb8ccb95c21236bad2a4d29
results/loco_table_rel.csv	974991860e46c040e6d82562267f738e5f8e88ef59547525eaec78800f21f1d0
results/overlap_test.json	f97abdbd845ae6640c9a616200608bf0a3ff3a4e30290260efbcf4430e2fcf3a
results/panel_coverage.csv	37ded335eabf606430d148ec35e049c842b260efe878e758c043e9cddf9cf345
results/reference_features.csv	08656183bebce926ccf5742c84b480f57996250f73434063bad849da01d6dd2d
results/route_decision.json	38e86fdf4a7d1a55380482a56d6f2e07f7e8be8cc42e8585c5e4f1a698ad59cc
results/stamps/00_build_global_panel.json	1ea794a5de1c7b14b143124b16635dabf1cb773b4436bcec1a063addd1adaf53
results/stamps/02_case_effects.json	c8995d389fbdc6857e4b54af8694822a0bb9beb14cfc921305fe66628d775b3a
results/stamps/03_feature_importance.json	25fcecd6f8a53f7bfec9ea796c879df25fbf27a1f0015ab12d9f9fe5f41e3584

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results/stamps/04_optimal_transport.json	75bbcbf3d18b1e72eb04e7bbb51c216a42a2e2a23d17e93df386315e6f8dac66
results/stamps/05_thailand_impact.json	6a70ef48b995ad1d78301439013d96c060d33db9ae20485f4097b801f4cb563b
results/thailand_detection.csv	e2c8a63960c0c40698323b15bb3a7f6b4204426bc09fe31df800bc18f75db81a
results/thailand_impact.csv	33d4e59038f44a4f0bf7f5cf933a787cdf7330f76f6e4f24fdc2e5a0ea316e00
results/thailand_scenario_draws.csv	63af966fccbbd179ac90a0173699f0b43b8cc89e6e9b8e657af139fbb4ada686
results/thailand_scenario_null_draws.csv	2a9fe6263944c82826f1be91e4774251d16b6a85e1710c3baae68c3a16b6f7b0
results/thailand_scenario_usage.csv	7b5a7a6a6a4a1e0d7c0aab18fb9313b9cbcf656935431acc7416eb8d1ba579c4
results/thailand_scenarios.csv	efc2bc397545ff1ca11c23b4c75889c8ff610fd27aec3dac3c3eb21e91540bf3
results/thailand_summary.json	b1da49df38fdec408947cda987125373bc15e7b79b7debe530f3374d6f3fe83f
results/thailand_target_cloud.csv	12d551bf826bf0f6ed90c2a5f35b542dfa3da7dd3980a2071de39f1ee46bbc7e
results/thailand_transport.json	9c5ffa0d7ba09252dcca0b201ce23f6eaadcec568d727dfbb4bfee6ff77e70f8
results/thailand_transport_draws.csv	1f26c87cda697821cf8f3284b5bd8229ee11724e1133911107d814aa9139b013
results/transport_plan.csv	9fe4c69ea3424e14623b3cb744157c742d341a62a633d1064783a10b72ef112b
results/transport_sensitivity.csv	12b3fd8b2c583cdccdfbdbd2c860fb2bfb7634d3eeb916c2e9255a41a26e4276
results/transport_usage.csv	5dd02ca670fc49960c1c56f8ee968b7426c4c3fe2dc65686abfbf185c7d34a86
results/transport_weights.csv	9092d93e833844924ca0d864af5aab0b1e4cace90f4406a8eeabeaa92e051f78
scripts/verify_wdi_files.py	06ba7bf129623320659241d6a26324391dd90fbb78a02ad59fb6db932b6665d5
src/dtt/__init__.py	a546f33411d27cb0ca1fde1d39482e392cc7eea5d2d5fccbe5a8805296477f89
src/dtt/appraisal.py	9603e845f9ca43d229c6771d95cc205acd603aabcecef3718ee5d13ce2deac2f
src/dtt/cases.py	381e11984e5ff0ebe78e6adb648e447ccbaf87aa416691bc30d47bbe623e3b7e
src/dtt/did.py	57422bd2f941be629f1479b18e6a486c16acde0e5d775a8b7a3c7cde0aea6be4

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src/dtt/effects.py	98822093671488fce84bf30253bd8285f04d3bb46b83b0469d4024cff2e8978b
src/dtt/impact.py	624f02a1686a6e0c384e403cb00b4afbfc51f07b2f03ae7f38225aba9aa9d5dc
src/dtt/importance.py	ca8daf3182898bc0001b908a4571940eca881e52f2de62ef3da5d109e64bab66
src/dtt/meta.py	2b5b58cd9ce5771553c4be2bbeaa2a387b6294b9ee70af7347efac727310be03
src/dtt/panel.py	703e0f381838e74c6d7d1541fa1598c702460a2a0b048050666e3aa051d39ae1
src/dtt/replication.py	fd1de9febf171e9bada1de739cf735543e89049297154b6cb93963e232e5d7c0
src/dtt/scm.py	d05224f19b58b1af8e5f6b023e00e536b49ca42ce4ed0bda04e75dc23eec562c
src/dtt/stata_compat.py	6da80499e41bf73036d3d15cd9d5d63571e37fcf7932d75fd130455b4026c0a3
src/dtt/thermal.py	620c08df01ea49da30b7ae6e0cd9eb08f68b457a6ba4e3e73b6fa5a0ac692220
src/dtt/transport.py	8ae373124faff0ad4fe6d1ead4454e735437908afd3f1d57908280a752054231
src/dtt/workflow.py	edd063674f6c2ddab45faebf3051a952c720248448eb567465ca26731f83543b
tests/conftest.py	d9f9b3ccdacfe4ccda6f2962f0895322838a414e21382dc3817bb31882cdb912
tests/handmade_panels.py	896e03eea13cf43c5ea30517b0fb0df780fd2bd736cec1998d1f86001e04e247
tests/handmade_world.py	d6019eaae092979d69fe1a386fb809966ca99b6b21317d6210a24df74eb7154b
tests/test_appraisal.py	0e212368d90a4306a3c1817f2eaf5f9a07a3f659adf2ba45cfdc0b80cc9fc175
tests/test_cases.py	2866b7ea4638394ccf746c0c05c0ced56f84b22c1962ed6d632d440a77effdf6
tests/test_did.py	f6a521e4540297471a07a4b0fdfc173154365fb6b813f14fbde61cda7254c780
tests/test_effects.py	51e07f359ef6022a63a1943d02c8c99c80ed7df96aa35beac0668a197e820bbf
tests/test_impact.py	f364ec3ea28d223538c2d2b883c74f7ff7a098ef086e66d34d27e5e0c69658cf
tests/test_importance.py	588713a0ab978188c68cf65ce415954b117c74566fb0346f69298181a66888f1
tests/test_meta.py	8aa967d38709cdcd792ee7d165f272e2cc416a0da6f5b7f078a312bbf3d890d3

File	SHA-256
tests/test_notebooks_static.py	29f9db1589869b5fc2de663a0e7cfa2152b7b84375a015b162dfcbc91f476855
tests/test_panel.py	932402cb360cde2c685b250759597f7f1f532535db6556287e29a6670243e465
tests/test_replication.py	e6b98fa4b74a11b64231c8686ad78b84fe08ed0d60b99f83793cbf386200bda2
tests/test_scm.py	f3e1a669417889eb085b2c07d14931a27fb2d27d1fb2ec8fb38fac0aecfed603
tests/test_se_conventions.py	b25a7e9c9b958034195b3d3f087b07044e8b9938275e606fdbf10f90135c71bb
tests/test_stata_compat.py	cde2b4cbd5db3650992e30745cb0b814944f7d4bc41fa59e1c2f7999791063d9
tests/test_thermal.py	676a38647bf634c81b4987f829a71fa8508dead9b02bd248933f0aa5912a7a73
tests/test_transport.py	3057f6015bffc45e4168eb8771c7aed32b6708dcf82c1c014f5aca13186569a0
tests/test_verify_wdi_files.py	d3104d0c7ad7ec44d1e1ab5472ffc7a6419901a20486616cbb84da52a3ef5739
tests/test_workflow.py	63a2fc5505b68accb79061ce10c0c79defa94a3a864eebf91422a3bfb2de9a72